Patient Profile

- Name: Mr. Saubhik Bhaumik
- Age: 27 years
- Gender: Male

Summary of Results

Liver function parameters, including bilirubin fractions, transaminases (AST, ALT), total protein, albumin, globulin, and albumin–globulin ratio, are within the provided reference ranges. The value for serum alkaline phosphatase is not reported, so this component cannot be assessed from the current data.

ML Result: Normal – the predictive model assesses the profile as within normal limits.

Detailed Test Explanations

SERUM BILIRUBIN (TOTAL)

- Patient result: 0.9 mg/dl
- Reference range: 0.2–1.2 mg/dl

Total bilirubin reflects the sum of direct (conjugated) and indirect (unconjugated) bilirubin and is a key marker of liver function, bile flow, and red blood cell breakdown.

- Elevated levels may indicate liver cell dysfunction (hepatitis, cirrhosis), bile duct obstruction (cholestasis), or increased red blood cell destruction (hemolysis).
- Low or low-normal values are generally not clinically concerning.

The patient’s total bilirubin is within the normal range, indicating no biochemical evidence of jaundice or significant bilirubin handling abnormality.

SERUM BILIRUBIN (DIRECT)

- Patient result: 0.2 mg/dl
- Reference range: 0–0.3 mg/dl

Direct (conjugated) bilirubin is produced when the liver processes unconjugated bilirubin and prepares it for excretion into bile.

- Elevated direct bilirubin often suggests cholestasis, biliary obstruction, or hepatocellular disease affecting bile excretion.

The patient’s direct bilirubin is within normal limits, suggesting no evidence of conjugated hyperbilirubinemia or overt cholestasis.

SERUM BILIRUBIN (INDIRECT)

- Patient result: 0.70 mg/dl
- Reference range: 0.2–1.0 mg/dl

Indirect (unconjugated) bilirubin represents bilirubin before hepatic conjugation.

- Elevated levels may be seen in increased red cell breakdown (hemolysis), impaired hepatic uptake/conjugation, or certain inherited conditions (e.g., Gilbert syndrome).

The patient’s indirect bilirubin is within normal limits, suggesting normal bilirubin production and hepatic handling.

SGPT (ALT)

- Patient result: 36 U/L
- Reference range: 13–40 U/L

Alanine aminotransferase (ALT) is an enzyme primarily found in liver cells. It is a sensitive marker of hepatocellular injury.

- High ALT typically reflects liver cell damage from causes such as viral hepatitis, fatty liver disease, alcohol-related injury, medications, or toxins.
- Very low ALT usually has limited clinical significance in otherwise healthy individuals.

The patient’s ALT is within the normal range, indicating no biochemical evidence of active liver cell injury.

SGOT (AST)

- Patient result: 32 U/L
- Reference range: 0–37 U/L

Aspartate aminotransferase (AST) is found in liver, heart, muscle, and other tissues.

- Elevated AST can indicate liver injury but may also reflect muscle damage, cardiac injury, or other systemic conditions.
- Interpretation often considers the AST/ALT ratio and absolute values.

The patient’s AST is within the normal range, supporting the absence of significant hepatocellular damage based on this data.

SERUM ALKALINE PHOSPHATASE

- Patient result: Not reported
- Reference range: Not provided

Alkaline phosphatase (ALP) is an enzyme present in bile ducts, bone, and other tissues.

- Elevated ALP may suggest cholestasis, bile duct obstruction, infiltrative liver disease, or increased bone turnover.
- Low ALP is less common and may be seen in certain nutritional or metabolic conditions.

Because no numerical value or reference range is provided, this parameter cannot be assessed from the current report and should be repeated if clinically indicated.

SERUM PROTEIN

- Patient result: 7.2 g/dl
- Reference range: 6.4–8.3 g/dl

Total serum protein includes albumin and globulins and reflects nutritional status, liver synthetic function, and immune protein levels.

protein levels.

- Low levels may indicate malnutrition, malabsorption, liver synthetic dysfunction, or protein loss via kidneys or gut.
- High levels may suggest chronic inflammation, infection, or certain blood protein disorders.

The patient’s total protein is within normal limits, indicating adequate overall protein status and no evident impairment of hepatic protein synthesis.

SERUM ALBUMIN

- Patient result: 4.7 g/dl
- Reference range: 3.5–5.2 g/dl

Albumin is the main plasma protein synthesized by the liver and is a marker of liver synthetic function, nutritional status, and chronic disease burden.

- Low albumin may be seen in chronic liver disease, nephrotic syndrome, protein-losing enteropathy, severe malnutrition, or systemic inflammation.
- High values are uncommon and usually of limited concern in isolation.

The patient’s albumin level is comfortably within normal range, supporting normal liver synthetic capacity and generally adequate nutrition.

GLOBULIN

- Patient result: 2.5 g/dl
- Reference range: 1.8–3.6 g/dl

Globulins include various proteins such as immunoglobulins and acute-phase reactants.

- Elevated globulin levels may reflect chronic inflammation, autoimmune disease, chronic liver disease, or plasma cell disorders.
- Low globulin may occur with immune deficiencies or protein losses.

The patient’s globulin level is within the normal range, suggesting no major disturbance in globulin production or loss.

A/G RATIO (Albumin/Globulin Ratio)

- Patient result: 1.88
- Reference range: 1.1–2.4

The albumin/globulin ratio compares albumin to globulin levels and helps refine the interpretation of total protein.

- A reduced ratio can be seen in chronic liver disease, autoimmune disorders, chronic infections, or increased globulin production.
- An elevated ratio may occur with low globulin states but, when within reference, is generally reassuring.

The patient’s A/G ratio lies well within the normal range, consistent with balanced albumin and globulin fractions and absence of significant chronic inflammatory or protein dysregulation evident on this panel.

Overall Interpretation

Taken together, the available liver function test parameters (bilirubin fractions, ALT, AST, total protein, albumin, globulin, and A/G ratio) are all within their stated reference ranges. This pattern does not suggest active hepatocellular injury, cholestasis, or impaired hepatic synthetic function at this time.

The absence of a reported alkaline phosphatase value is a limitation; however, based on the parameters provided, there is no biochemical evidence of liver dysfunction or jaundice.

The predictive model's assessment is flagged as **Normal**, and it **assesses the profile as within normal limits**, which is concordant with the laboratory data presented.

Recommendations

1. **Clinical correlation**
    - Discuss these results with a healthcare provider, especially if there are symptoms such as fatigue, abdominal pain, jaundice, dark urine, pale stools, itching, or unintended weight loss.
    - If there is a history of liver disease, heavy alcohol use, metabolic risk factors, or hepatotoxic medication use, periodic monitoring of liver function tests may be appropriate.
2. **Follow-up on missing parameter**
    - Because the serum alkaline phosphatase value is not reported, consider repeating a full liver function panel (including ALP and, if appropriate, GGT) if there are risk factors or symptoms suggestive of biliary or liver disease.
3. **Lifestyle and liver health**
    - Limit or avoid alcohol intake; if alcohol is consumed, keep within recommended low-risk limits.
    - Maintain a healthy body weight through a balanced diet and regular physical activity to reduce the risk of fatty liver disease.
    - Avoid unnecessary over-the-counter medications, supplements, or herbal products that may affect the liver; use all medications as prescribed.
    - Stay hydrated and follow a diet rich in vegetables, fruits, whole grains, and lean protein while minimizing highly processed and very high-sugar foods.
4. **Infection prevention and vaccination**
    - If not already done, discuss vaccination for hepatitis A and B with a healthcare provider based on personal risk profile.
    - Practice safe behaviors to reduce the risk of viral hepatitis (e.g., safe sex practices, avoiding sharing needles or sharp instruments).
5. **When to seek prompt medical attention**
    - Seek urgent medical care if symptoms such as yellowing of skin or eyes, severe abdominal pain, confusion, marked fatigue, or significant darkening of urine and lightening of stools develop.

Disclaimer

This report is for informational and educational purposes based on the provided laboratory data and predictive model output. It does not replace a physical examination, detailed medical history, or personalized advice from a licensed healthcare professional. All results should be interpreted and confirmed by your treating clinician, who can provide diagnosis and management tailored to your specific clinical context.